Creatine

Files for a bioinformatic analyses on creatine effects on kidney tissues.

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Creatine review

Running Title

Creatine effects on kidney tissues and renal function: new insights to old questions.

First Steps

Creatine-related proteins on Stitch Database

Our first step was to use the term "*creatine*" in a search using the Stitch Database (PUBMED), which list the known and predicted interactions between chemicals and proteins. The search was limited to the organism *Homo sapiens*. We found the proteins listed in the file stitch_interactions-1st.tsv. Then, using the advanced settings, we restricted the search for integration of RNA-Seq expression data from kidney tissues, using the Human Protein Atlas database. The file stitch_interactions-2nd-ProteinAtlas.tsv contain the updated list, where the proteins AKT1, AKT2 and AKT3 appeared as a tissue-specific interactions. We also used the restriction from Tissues Database, using kidney as a term stitch_interactions-3rd.tsv.

First observations

- The genes GATM and SLC6A8 were present in all three above searches.
- The genes from AKT family were present using expression data from the Human Protein Atlas.
- The genes PSMB5, PSMD3, SGK1 and PRPS1 appeared when using expression data from the Tissues Database.

The first suggested protein list is (Stitch-1st):

```
SLC2A4
IGF1
GATM
SLC6A8
AKT1
AKT2
AKT3
PSMB5
PSMD3
SGK1
PRPS1
```

Since our aim is to review gene expression and protein alterations induced by creatine that are specific to renal tissues, we excluded the genes that are expected to interact with creatine, such as CK and CKMT. Interestingly, these genes were not found in the 3rd list (from Tissues database).

Using the Comparative Toxicogenomics Database

Our second step was to perform searches using the Comparative Toxicogenomics Database. As a result, we obtained a list with 44 entries CTD_gene_results_20240409072043.tsv. After filtering the repetitive entries, we obtained the following gene list:

```
CKB
CKBE     Locus
CKBP1    Pseudogene
CKM
CKMT1A
CKMT1B
CKMT2
SLC16A12
SLC6A10P     Pseudogene
SLC6A10PB    Pseudogene
SLC6A8
```

Interestingly, three pseudogenes and one *locus* were retrieved. The informations from each gene symbol were obtained from GeneCards database. The validated symbols were the following ones CTD-genes-list-Edited-1st.tsv:

```
CKB
CKM
CKMT1A
CKMT1B
CKMT2
SLC16A12
SLC6A8
```

After that, we used the batch query tool from CTD, using NCBI gene symbols from the two above lists as a input. Then we got the following results:

- For Stitch-1st list:
    - Curated Gene/diseases associations: CTD_gene_diseases_curated_1713265713318.tsv.
    - Inferred Gene/diseases associations: CTD_gene_diseases_inferred_1713266126277.tsv.
    - Curated Gene/pathways associations: CTD_gene_pathways_curated_1713266155612.tsv.
    - Chemical/Genes interactions at the expression level: CTD_gene_cgixns_1713266196128.tsv.
    - Curated Gene/Chemicals interactions: CTD_gene_chems_curated_1713267587656.tsv.
- For CTD-genes-list-Edited-1st.tsv:
    - Curated Gene/diseases associations: CTD_gene_diseases_curated_CTD-list.tsv.
    - Inferred Gene/diseases associations: CTD_gene_diseases_inferred_CTD-list.tsv.

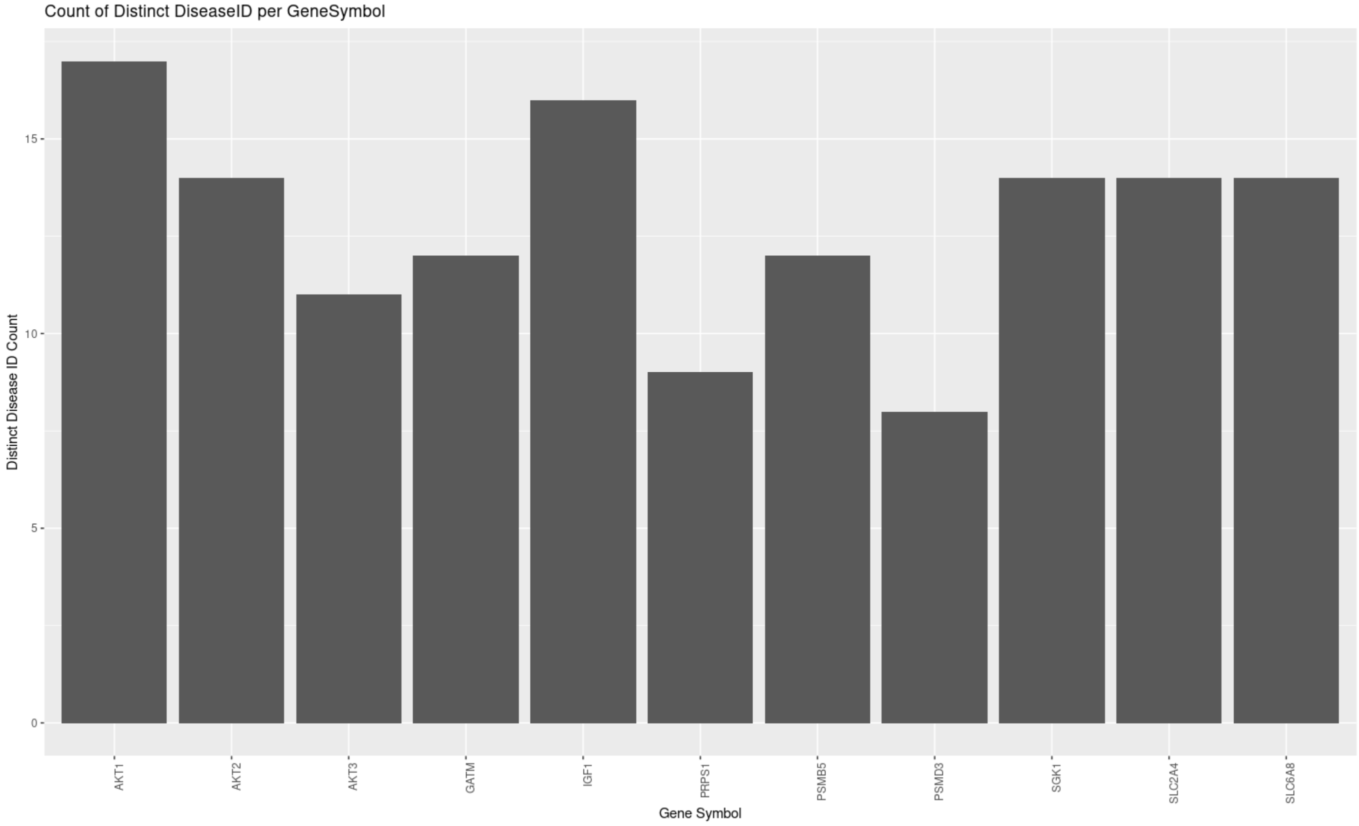
- Curated Gene/pathways associations: CTD_gene_pathways_curated_CTD-list.tsv.
- Chemical/Genes interactions at the expression level: CTD_gene_cgixns_CTD-list.tsv.
- Curated Gene/Chemicals interactions: CTD_gene_chems_curated_CTD-list.tsv.

Filtering CTD's results

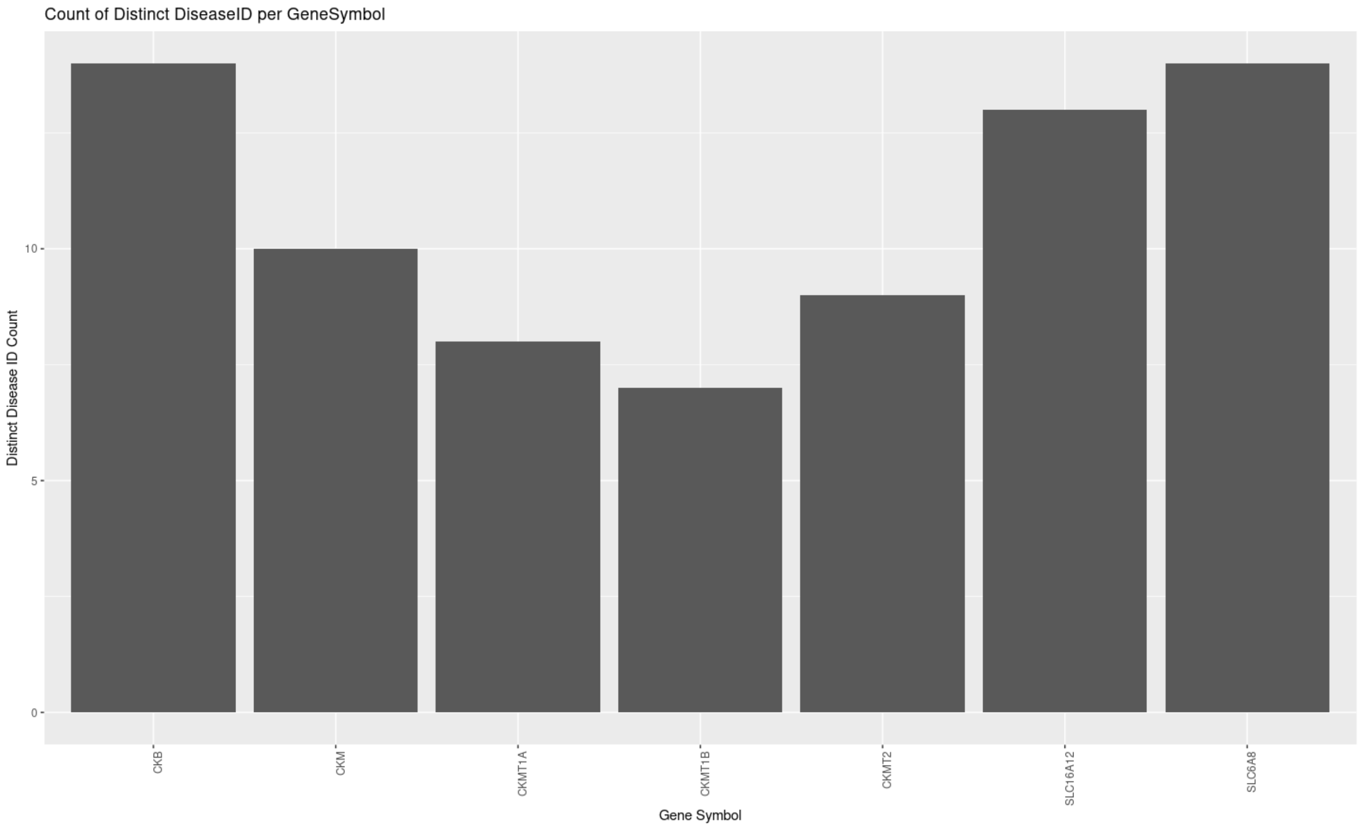
We started filtering the inferred gene/diseases associations, since the curated ones are enriched to genetic diseases (inborn). For the Stitch-1st and CTD lists, we obtained a file with 153952 and 50868 rows, respectively. Then, we filtered the results using the presence of the terms `kidney` and `renal`, obtaining 2596 and 818 rows, for Stitch and CTD gene lists. The higher number of entries in the Stitch list is probably associated with the initial filtering process for expression in kidney tissues. The following table describes the number of entries per gene, in each list:

Stitch		CTD	
Gene	**Number of entries**	**Gene**	**Number of entries**
SLC2A4	283	CKB	214
IGF1	431	CKMT1A	39
GATM	172	CKMT1B	37
SLC6A8	157	CKMT2	90
AKT1	691	SLC16A12	114
AKT2	192	SLC6A8	157
AKT3	111		
PSMB5	119		
PSMD3	84		
SGK1	268		
PRPS1	109		

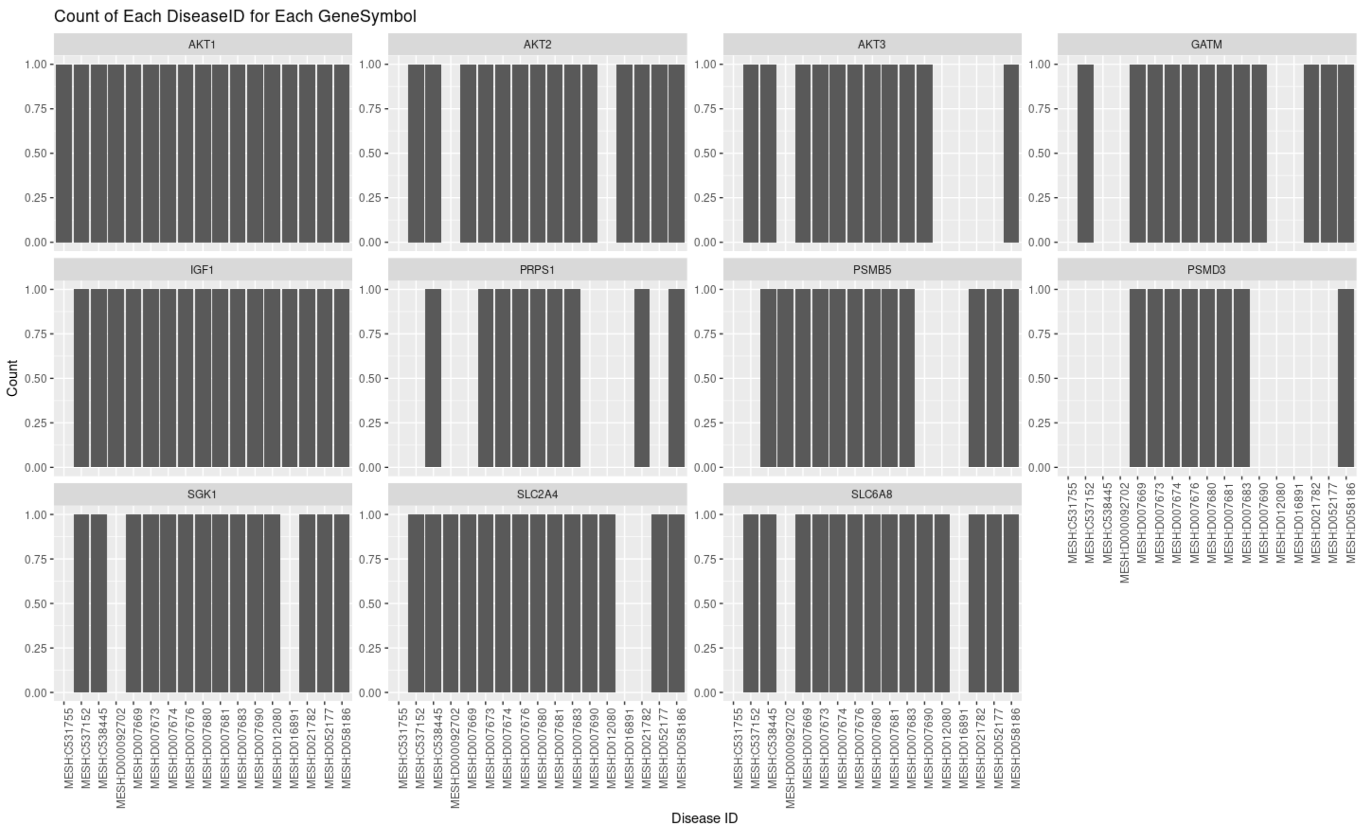
From the filtered files, we obtained the unique DiseaseIDs, and unique DiseaseIDs/gene.



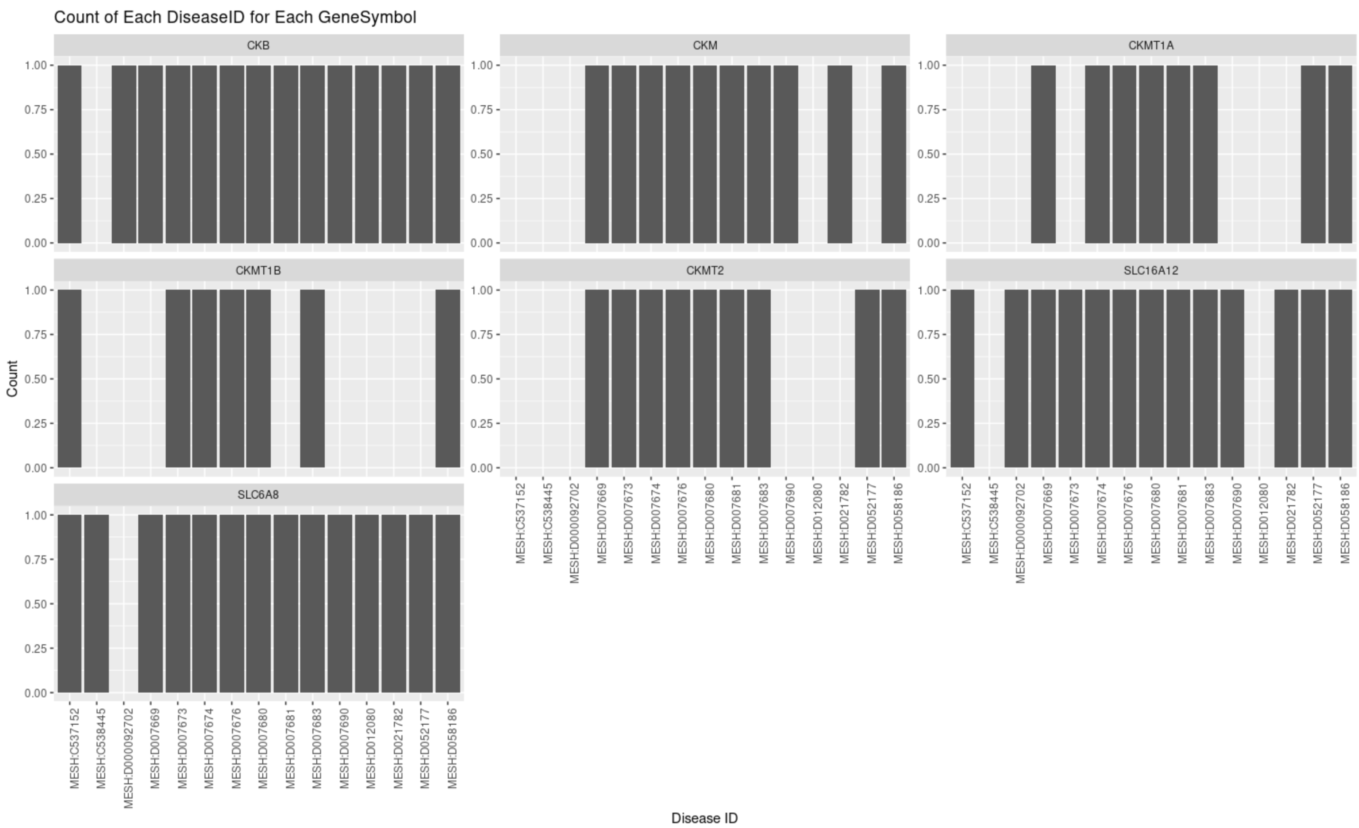
Stitch Genes with associated DiseaseIDs related to Kidney/Renal diseases



CTD Genes with associated DiseaseIDs related to Kidney/Renal diseases



Presence/Absence of DiseaseID per gene from the Stitch-list



Presence/Absence of DiseaseID per gene from the CTD-list

Here are the number of DiseaseIDs and counts, common to the two genes lists:

```
DiseaseID
MESH:D007676       77   Kidney Failure, Chronic
MESH:D007674       77   Kidney Diseases
MESH:D007680       77   Kidney Neoplasms
MESH:D007683       77   Kidney Tubular Necrosis, Acute
MESH:D058186       77   Acute Kidney Injury
MESH:D007681       66   Kidney Papillary Necrosis
MESH:D007673       66   Kidney Cortex Necrosis
MESH:D007669       60   Kidney Calculi
MESH:D052177       40   Kidney Diseases, Cystic
MESH:C537152       32   Hypomagnesemia 2, renal [Supplementary Concept]
MESH:D007690       32   Polycystic Kidney Diseases
MESH:D021782       32   Multicystic Dysplastic Kidney
MESH:D012080       10   Chronic Kidney Disease-Mineral and Bone Disorder
MESH:C538445        9   Clear-cell metastatic renal cell carcinoma
[Supplementary Concept]
MESH:D000092702     8   Chronic Kidney Diseases of Uncertain Etiology
```

The list from the Stitch database has these unique DiseaseID counts:

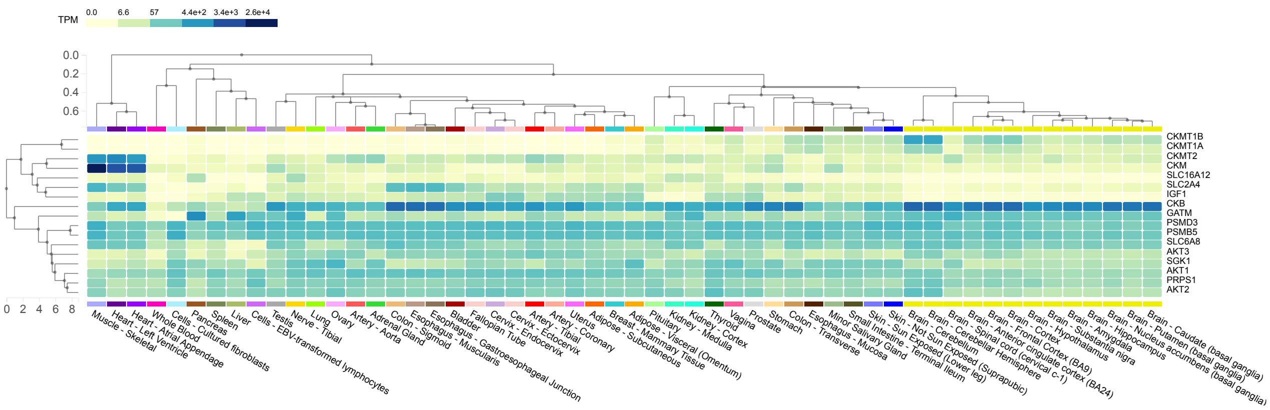
```
DiseaseID
MESH:D016891    3   Polycystic Kidney, Autosomal Dominant
MESH:C531755    1   Kidney disorder involving deposition of calcium and oxalate
or phosphate in the renal tubules [Supplementary Concept]
```

Gene Expression

We merged the two above lists into one.

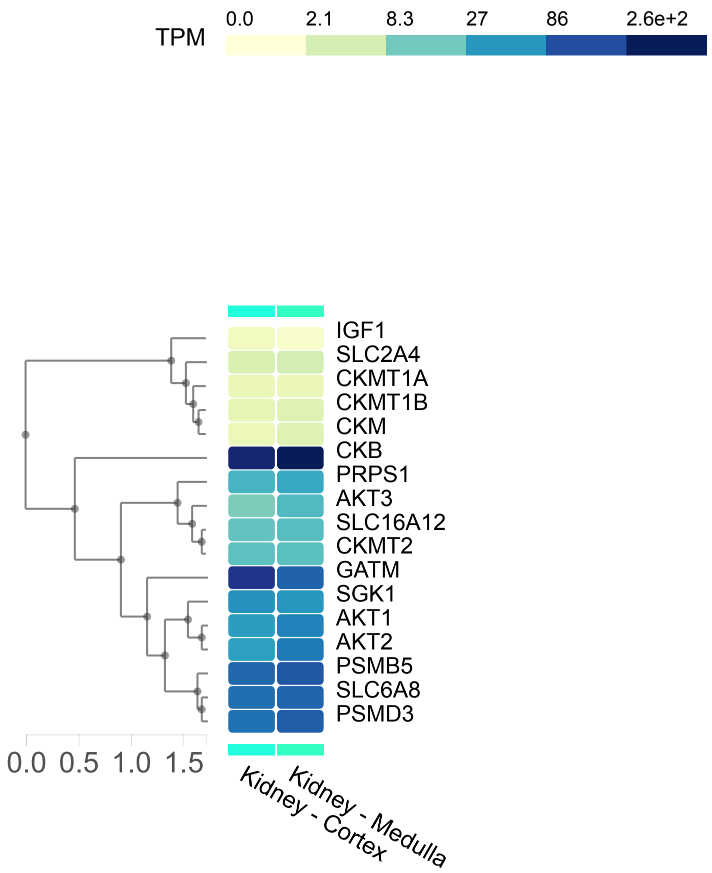
```
SLC2A4
IGF1
GATM
SLC6A8
AKT1
AKT2
AKT3
PSMB5
PSMD3
SGK1
PRPS1
CKB
CKM
CKMT1A
CKMT1B
CKMT2
SLC16A12
```

The first step was to retrieve a general expression panel of these genes, for all tissues. For this we used the tool `Multi-Gene Query` of the GTEx Portal. The figure below describes the relative expression of these genes in transcripts per million (TPM).



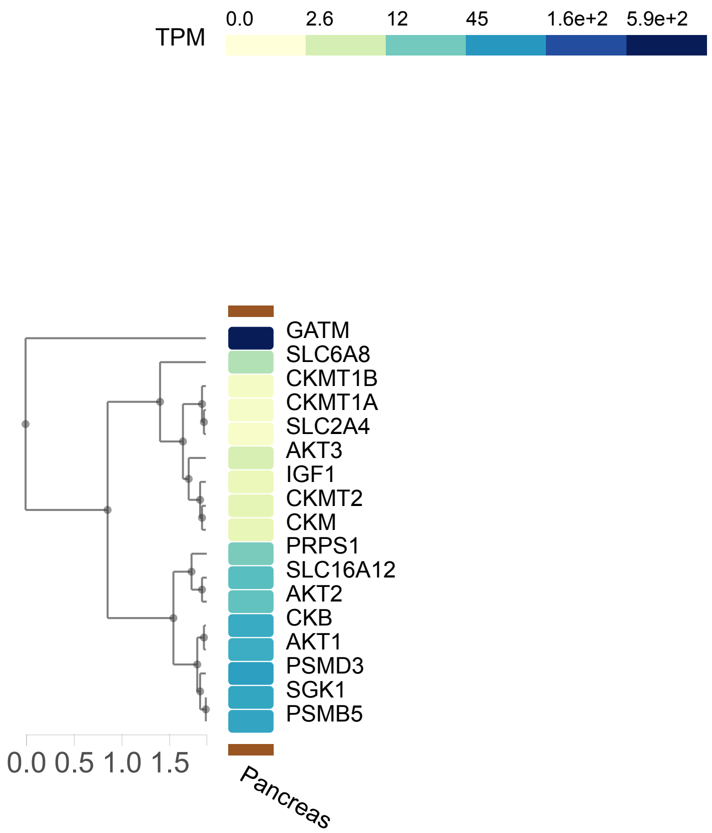
The general expression panel for all tissues of the 17 creatine-related genes used in the present study

We also highlighted the specific expression for kidney tissues:



Kidney tissues expression levels of the 17 creatine-related genes used in the present study

And from pancreas tissues:



Pancreas expression levels of the 17 creatine-related genes used in the present study

Important Information

As highlighted above, we noticed the following results from GTex database:

CellTypes	P-value	adjusted P-value	OddsRatio	C
Pancreas Female 30-39 Up	0.00320528834894988	0.0293818098653739	27.05442176870748	15
Pancreas Female 20-29 Up	0.00320528834894988	0.0293818098653739	27.05442176870748	15
Pancreas Female 40-49 Up	0.00320528834894988	0.0293818098653739	27.05442176870748	15
Pancreas Female 60-69 Up	0.00320528834894988	0.0293818098653739	27.05442176870748	15
Pancreas Female 50-59 Up	0.00320528834894988	0.0293818098653739	27.05442176870748	15
Pancreas Male 20-29 Up	0.00320528834894988	0.0293818098653739	27.05442176870748	15

- *Why do pancreas cells have these genes upregulated?*
- *In which conditions did these increased expression occur?*
- *Is it probable that a creatine-induced high expression of GATM and SLC16A12 is associated with the observed improvement of creatine-supplmented streptozotocin-induced diabetic rats? How?*

  Generative-IA search (From Copilot and Chat-GPT):

  > *Just to give us some ideas in an short text. Not for use in final texts and the listed references should be reviewed.*

Query: *Write something about the importance of the genes GATM and SLC16A12, as well as their respective proteins, in pancreatic tissues or pancreatic cancer.* **Combined answer:** *GATM (Glycine Amidinotransferase) and SLC16A12 (Solute Carrier Family 16 Member 12) are two genes that have garnered attention in recent years due to their potential significance in pancreatic tissues and pancreatic cancer.*

GATM Gene and Protein: *GATM encodes the enzyme Glycine Amidinotransferase, which plays a crucial role in creatine biosynthesis. Creatine is essential for providing energy to cells, particularly those with high energy demands like muscle cells. While initially identified for its role in muscle metabolism, emerging research suggests that GATM may also be involved in pancreatic tissue function. Studies have indicated that GATM expression may be altered in pancreatic cancer cells, although the exact mechanisms and implications of this alteration are still being elucidated. Understanding the role of GATM in pancreatic tissues could provide insights into both normal pancreatic physiology and the pathological processes underlying pancreatic cancer.* **SLC16A12 Gene and Protein:** *SLC16A12 belongs to the Solute Carrier Family of transporters and is involved in the transport of metabolites across cell membranes. Specifically, SLC16A12 encodes a monocarboxylate transporter involved in the transport of monocarboxylates such as lactate and pyruvate. While the role of SLC16A12 in pancreatic tissues is less well-characterized compared to GATM, there is evidence to suggest its involvement in various cancers, including pancreatic cancer. Dysregulation of monocarboxylate transporters like SLC16A12 can influence cellular metabolism, which is a hallmark of cancer cells.*

GATM (Glycine Amidinotransferase): *GATM plays a crucial role in the synthesis of guanidinoacetic acid (GAA), a metabolite associated with pancreatic ductal adenocarcinoma (PDAC) liver metastasis[1]>[6]. High expression of GATM has been positively correlated with advanced N stage in PDAC[1][6]. Knockdown of GATM significantly reduced the intracellular level of GAA, suppressed epithelial-mesenchymal transition (EMT), and inhibited PDAC liver metastasis[1][6]. These findings suggest that GATM-mediated de novo GAA synthesis >promotes PDAC metastasis[1][6].*

SLC16A12 (Solute Carrier Family 16 Member 12): *The SLC16A family members, including SLC16A12, play crucial roles in tumorigenesis and tumor progression[2][1]. However, the exact role of SLC16A12 in human pancreatic cancer remains unclear[2][1]. The SLC16A family of monocarboxylate transporters (MCTs) is known to play an important role in cell metabolism, such as aerobic glycolysis and pH homeostasis[2][1]. Current research progress on the MCT family, including SLC16A12, is mainly based on the findings of MCT1-4[2][1]. These proteins exhibit broad substrate specificity and are also involved in the transport of other monocarboxylic metabolites[2][1]. Please note that while these genes and their proteins have been associated with pancreatic cancer, the exact mechanisms and their potential as therapeutic targets are still under investigation. For more detailed information, please refer to the respective scientific research articles[2][1][1][6].*

Understanding the significance of GATM and SLC16A12 in pancreatic tissues and pancreatic cancer could have several implications:

- *Biomarkers: These genes and their respective proteins could serve as potential biomarkers for pancreatic cancer diagnosis, prognosis, or treatment response.*
- *Therapeutic Targets: Targeting the pathways involving GATM and SLC16A12 could offer novel therapeutic strategies for pancreatic cancer treatment.*
- *Molecular Mechanisms: Investigating the molecular mechanisms by which GATM and SLC16A12 contribute to pancreatic cancer development could provide insights into the underlying biology of the disease, potentially leading to the identification of new therapeutic targets or diagnostic tools.*

Overall, while further research is needed to fully elucidate the roles of GATM and SLC16A12 in pancreatic tissues and pancreatic cancer, their importance in cellular metabolism and potential implications for cancer biology make them intriguing candidates for further investigation in the field of oncology.

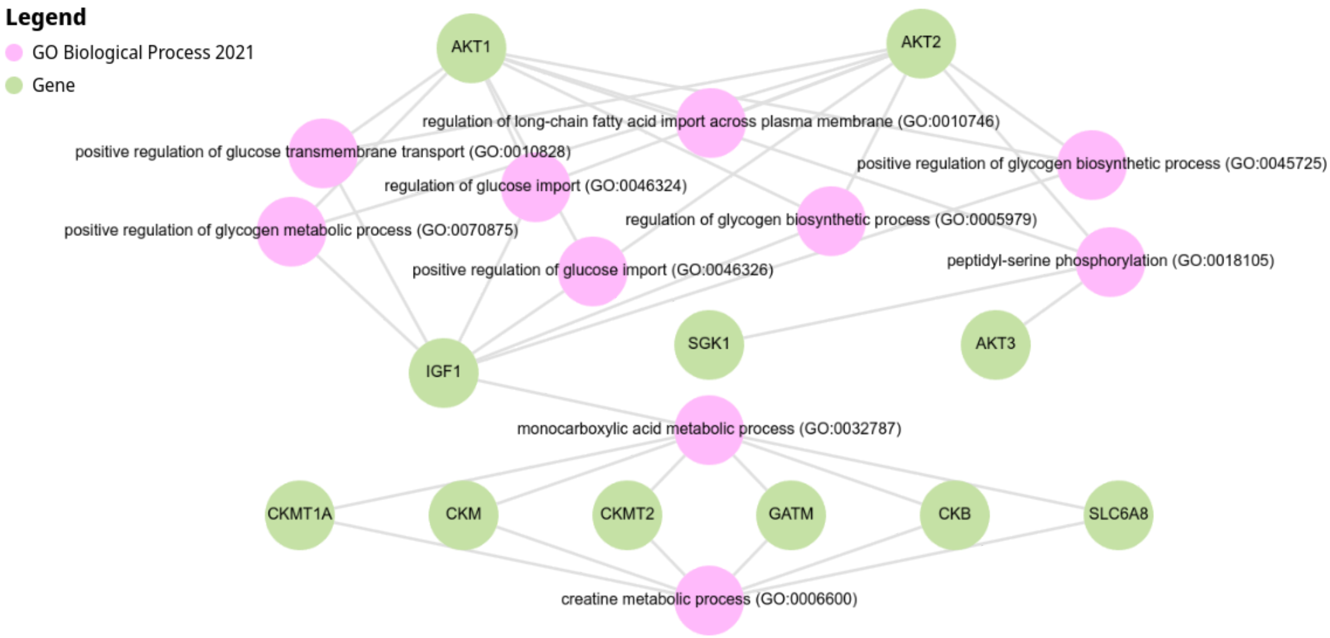
References: [1]: https://www.nature.com/articles/s41598-020-64356-y.pdf. [2]: https://www.mdpi.com/2072-6694/10/4/103. [3]: https://www.medrxiv.org/content/10.1101/2024.03.03.24303664v1.full.pdf. [4]: https://columbiasurgery.org/pancreas/genetics-pancreatic-cancer. [5]: https://link.springer.com/article/10.1007/s11605-022-05553-0. [6]: https://jeccr.biomedcentral.com/articles/10.1186/s13046-023-02698-x. [7]: https://www.nature.com/articles/s41392-023-01662-7.pdf. [8]: https://www.nebraskamed.com/cancer/pancreatic/the-role-that-genes-play-in-pancreatic-cancer. [9]: https://doi.org/10.3390/cancers10040103.

Gene enrichment analyses

We also used the final list as input for a gene enrichment analysis (GEA), using enrichR and enrichR-KG.

Gene Ontologies (GO)

Our first analysis was against the GO dataset to verify the terms associated to these 17 genes. The resulted network from the top ten P-value rated results is described in the figure below:



From the above figure one can note that the only direct integration of the creatine metabolism to all other proccess is the gene IGF1, which also participates in several carbohydrate-metabolism-related terms. Increasing the number of results to the top 30 rated results did not retrieved any other gene integration between creatine metabolism and other terms.

Cell-types analysis

The first filter used was CellTypes enrichment analysis. As a selection criterion, we chose the databases where the words "kidney" or "renal" were present with associated terms (P-value<0.05), from the all tested databases. Using the `.tsv` tables as input, we executed the following command:

```
awk -F'\t' '$1 ~ /Kidney/ && $3 < 0.05 {print $1, $3, $4, $7, $8, $9}' listName
```

We also searched against Kidney cell lines:

- **ARCHS4 cell-lines database**: 293F, FLPIN TREX 293, and HEK293.

Only three databases retrieved results within the selection criteria. The table below describes the terms and their associated values:

Database	CellTypes	P-value	adjusted P-valu
CellMarker 2024	Collecting Duct Intercalated Cell Kidney Mouse	0.003593	0.07843
HuBMAP ASCTplusB augmented 2022	Kidney Interstitial fibroblast - Kidney	8.808317003142788E-5	0.00130501787098
	Connecting Tubule Intercalated Cell Type A - Kidney	0.0033964046824717384	0.022655611815195
	Kidney Outer Medulla Collecting Duct Intercalated cell - Kidney	0.0034612740273215416	0.022655611815195
MoTrPAC 2023	T59-Kidney Consensus	0.0027283289091587022	0.02432356873463
	T59-Kidney Female 4W Up	0.004243143988996933	0.02670209257768
ARCHS4 cell-lines database	HEK293	0.04417020235184285	0.88573706595211
ProteomicsDB_2020	Renal SN-12C BTO:0004221 P0001751	0.04134985878495573	0.44706319361150

Database	CellTypes	P-value	adjusted P-valu
ARCHS4 Tissues	KIDNEY (BULK TISSUE)	2.952357524640197E-4	0.025095038959441
GTexGTEx Tissue Expression Up	GTEX-XPVG-0526-SM-4B65N kidney male 50-59 years	0.03125629534977868	0.2642102731529

Pathways and regulation

For the pathways enrichment analysis we used the results of just three databases: Reactome, Elsevier Pathway Collection, and KEGG. We also retrieved the analyses against kinases-regulation and PPI Hub proteins datasets in order to estimate the general regulation of the proteins coded by the creatine-related genes from our list. The inclusion criteria was the same for all analysis: the first 10 results, ordered by P-value<=0.05.

ReactomeDB

It generated the following results:

Term	P-value	adjusted P-value	OddsRatio
Creatine Metabolism R-HSA-71288	2.9184899635801106E-17	7.033560812228066E-15	2724.40909090909
Metabolism Of Amino Acids And Derivatives R-HSA-71291	2.3490852446193955E-10	2.8306477197663716E-8	49.0062421972534
Regulation Of PTEN Stability And Activity R-HSA-8948751	2.173230381584124E-9	1.7458284065392463E-7	133.87768817204
Cyclin E Associated Events During G1/S Transition R-HSA-69202	6.09405957169532E-9	3.3201285234231814E-7	107.716450216450
Cyclin A:Cdk2-associated Events At S Phase Entry R-HSA-69656	6.888233451085438E-9	3.3201285234231814E-7	104.97890295358
KEAP1-NFE2L2 Pathway R-HSA-9755511	1.6664329868330147E-8	6.693505830445942E-7	87.228070175438
Regulation Of TP53 Degradation R-HSA-6804757	2.0684898383664146E-8	6.983015044641323E-7	191.836538461538

Term	P-value	adjusted P-value	OddsRatio
Regulation Of TP53 Expression And Degradation R-HSA-6806003	2.3180132928269954E-8	6.983015044641323E-7	186.013986013986
RUNX2 Regulates Genes Involved In Cell Migration R-HSA-8941332	2.8485265782063636E-8	7.627721170530374E-7	856.2
Transcriptional Regulation By RUNX2 R-HSA-8878166	4.0042433537616845E-8	9.65022648256566E-7	72.620614035087

The first two terms were expected, since the gene list was originally retrieved using their interaction with creatine, and its overlap with the amino acid metabolism. Interestingly, the other eigth terms relate to processes involved in the regulation of the Cell Cycle, including the degradation and stability of the known tumor suppressor genes TP53 and PTEN. We also observed the overlap of the genes PSMB5, AKT2, AKT1, AKT3, PSMD3, and SGK1, with several retrieved terms. The integration of amino acid metabolism with some Glycolysis' reactions can explain this relationship with the cell cycle control. PKM2, a glycolytic enzyme, phosphorylates and activates ERK1/2, crucial for cell proliferation. Another glycolytic enzyme, PFKFB3, promotes cell proliferation by stimulating glycolytic ATP production and modulating the expression of cell cycle regulators Kalucka et al. 2015. The involvement of CK in mitosis regulation is well established in the scientific literature. In addition, previous studies report contrasting roles of the creatine metabolism in tumor cells. Both up and downregulation of CK may impair cell viability and induce cell death. The effect seem to depend on the nature of the tumor Yan et al. 2016. We can work with the following questions:

- How is the expression of crucial cell cycle regulators in kidney tissue samples from supplemented creatine patients (or murine models)? Is there any correlation with creatine supplementation?

- How do these levels compare with the ones observed for kidney cancer samples or kidney cancer-cell lines?

> *We can also apply the same questions using pancreas samples.*

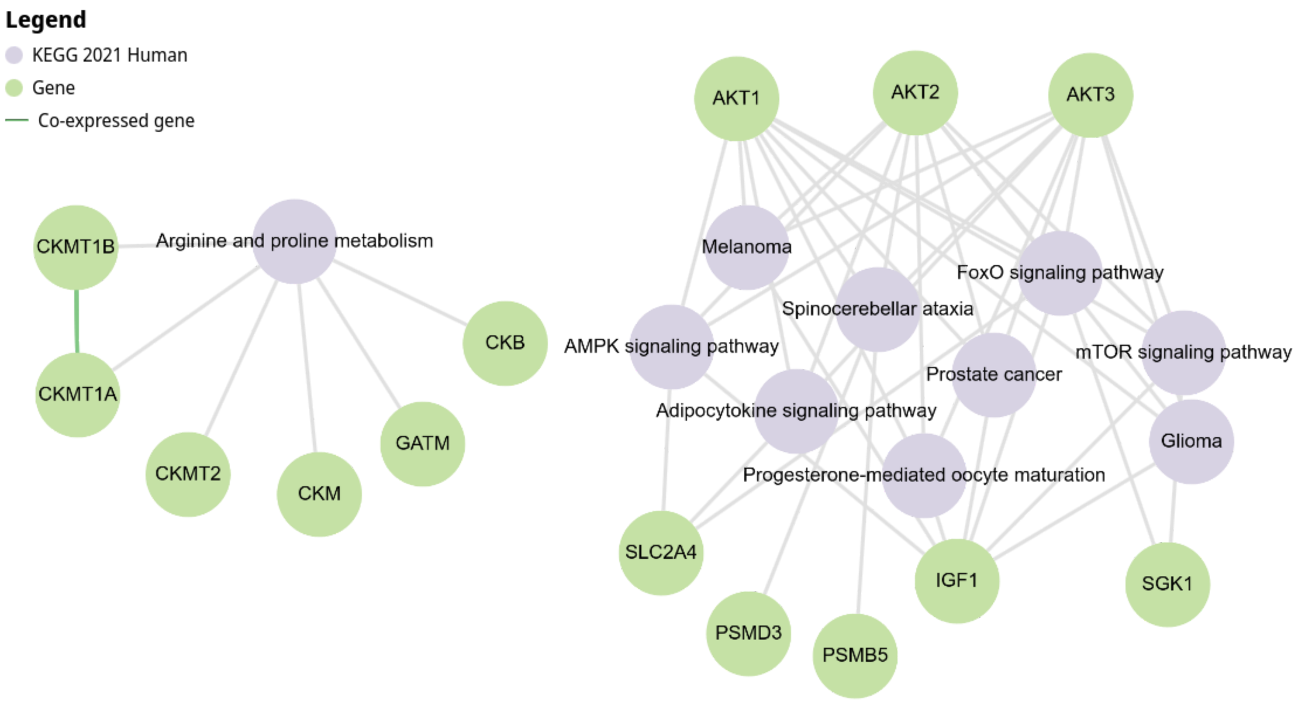
Kyoto Encyclopedia of Genes and Genomes (KEGG)

From KEGG, we got the following results:

Term	P-value	adjusted P-value	OddsRatio
Arginine and proline metabolism	2.167863990086027E-12	2.471364948698071E-10	247.17768595041
FoxO signaling pathway	8.206604458116714E-10	4.6777645411265273E-8	86.65309090909
AMPK signaling pathway	4.1762472319220094E-8	1.5869739481303636E-6	71.98550724637
Spinocerebellar ataxia	1.005795294602024E-7	2.8665165896157683E-6	59.91847826086
mTOR signaling pathway	1.4562974080079344E-7	3.3203580902580907E-6	55.46420581655
Adipocytokine signaling pathway	2.9841568453594475E-7	5.66989800618295E-6	94.28639053254
Melanoma	3.545726530398664E-7	5.774468920934967E-6	90.1131221719
Glioma	4.182518410872301E-7	5.9600887354930285E-6	86.29252437703
Prostate cancer	1.1787217874989481E-6	1.4930475974986676E-5	65.80645161290
Progesterone-mediated oocyte maturation	1.331899620230903E-6	1.4943595273294845E-5	63.74038461538

Other results within the same P-Value<=0.05 criterion correlate with insulin resistance and signaling metabolism, renal cell carcinoma, type II diabetes mellitus, central carbon metabolism in cancer, etc. However, the involvement of AKT genes in several metabolic processes explains the high number of associated terms.

The graph below represents the top ten results from KEGG terms:



The KEGG classification did not retrieve any integration between genes of the creatine metabolism process and all the other retrieved process. The green lines represent the genes that are co-expressed.

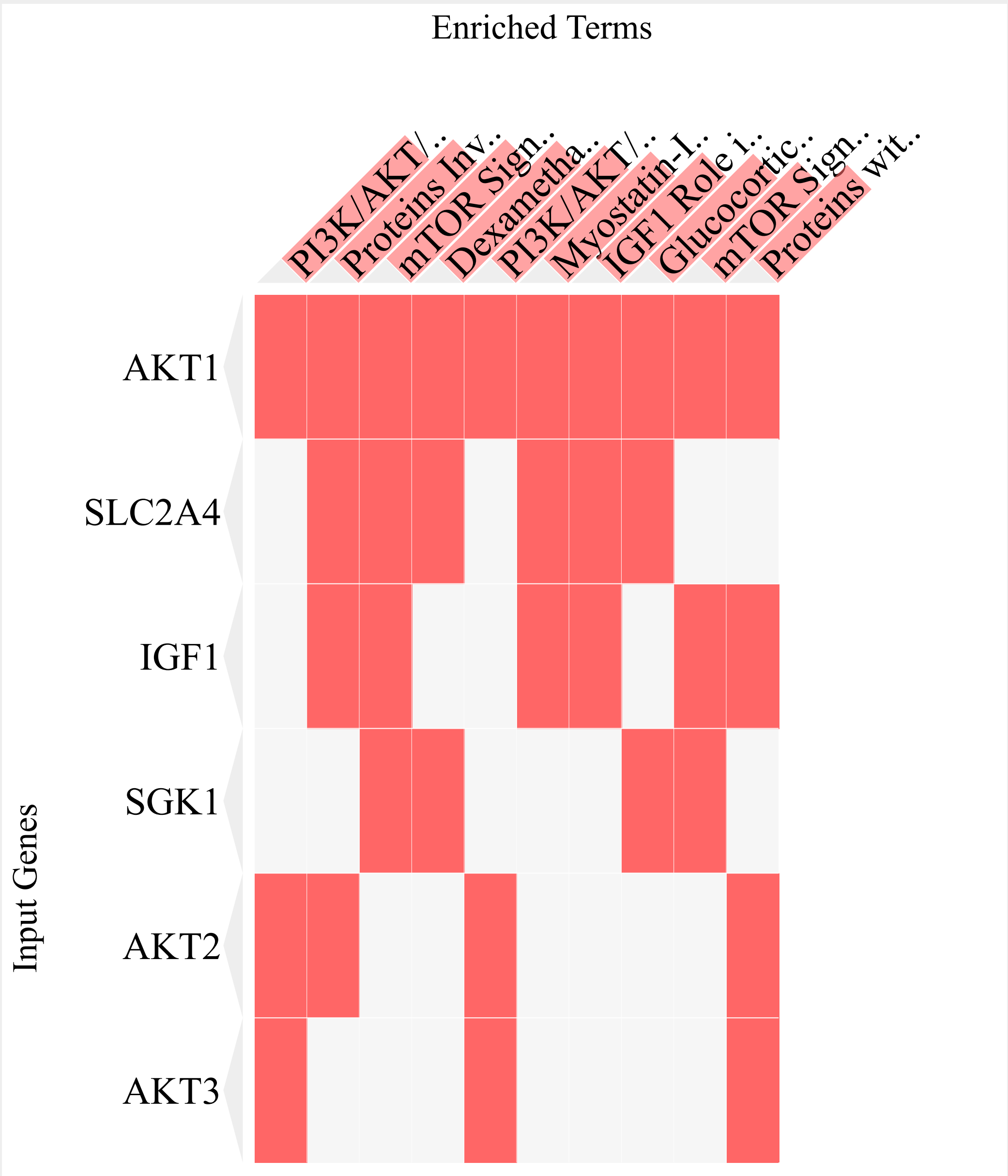
Elsevier Pathway Collection

The results were similar to the previous ones. Again, overlap of the genes from AKT family.

Term	P-value	adjusted P-value	OddsRatio
PI3K/AKT/MTOR Signaling Activation in Cancer	3.442744264155221E-7	9.138563669603297E-5	305.647959183
Proteins Involved in Insulin Resistance	5.707296999774423E-7	9.138563669603297E-5	79.54445554445
mTOR Signaling	5.707296999774423E-7	9.138563669603297E-5	79.54445554445
Dexamethasone Induced Diabetes	6.719532110002424E-7	9.138563669603297E-5	237.6785714285
PI3K/AKT/MTOR Signaling Activation by Blocking of Tumor Suppressors	1.15959377718675E-6	1.261638029579184E-4	194.4253246753
Myostatin-IGF1 Crosstalk in Skeletal Muscles in Muscular Dystrophies	2.4915350814739284E-6	2.1287904870031856E-4	147.4433497536
IGF1 Role in Muscle Hypertrophy	2.7392524648938054E-6	2.1287904870031856E-4	142.5214285714
Glucocorticoids in Insulin Resistance in Skeletal Muscles	3.2825551383464033E-6	2.2321374940755543E-4	133.6004464285
mTOR Signaling Hyperfunction	8.615629850848899E-6	5.207669598735334E-4	94.94285714285

Term	P-value	adjusted P-value	OddsRatio
Proteins with Altered Expression in Cancer-Associated Sustaining of Proliferative Signaling	1.2331668345829871E-5	6.708427580131449E-4	35.6491228070

The clustergram also evidences that AKT1 is involved in all significant associated terms:



Regulation by kinases

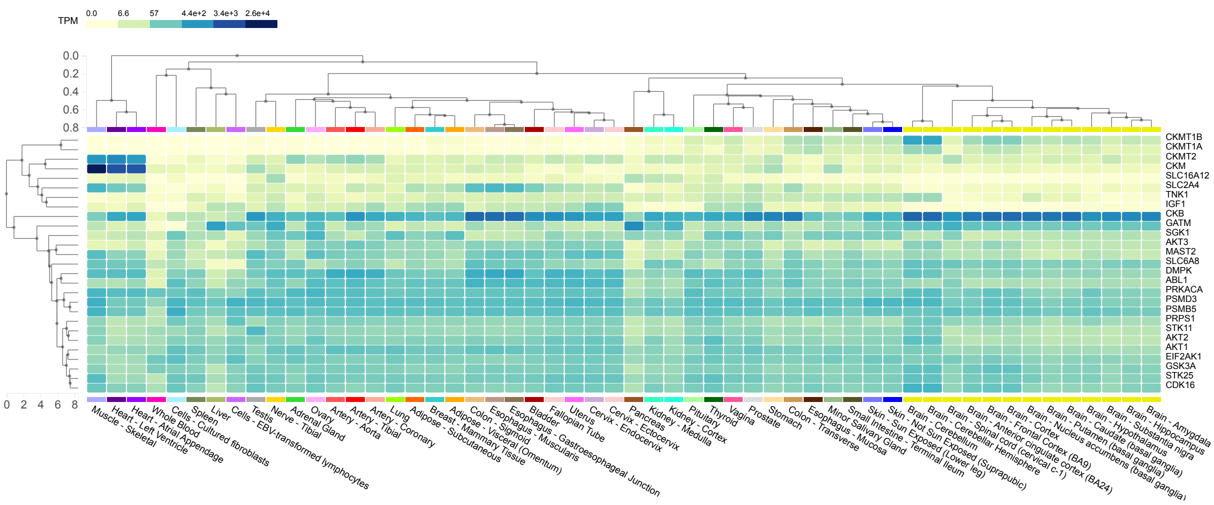
We chose three datasets to investigate regulation by kinases.

For the ARCHS4_Kinases dataset, which describe coexpression with several kinases, we obtained the following results:

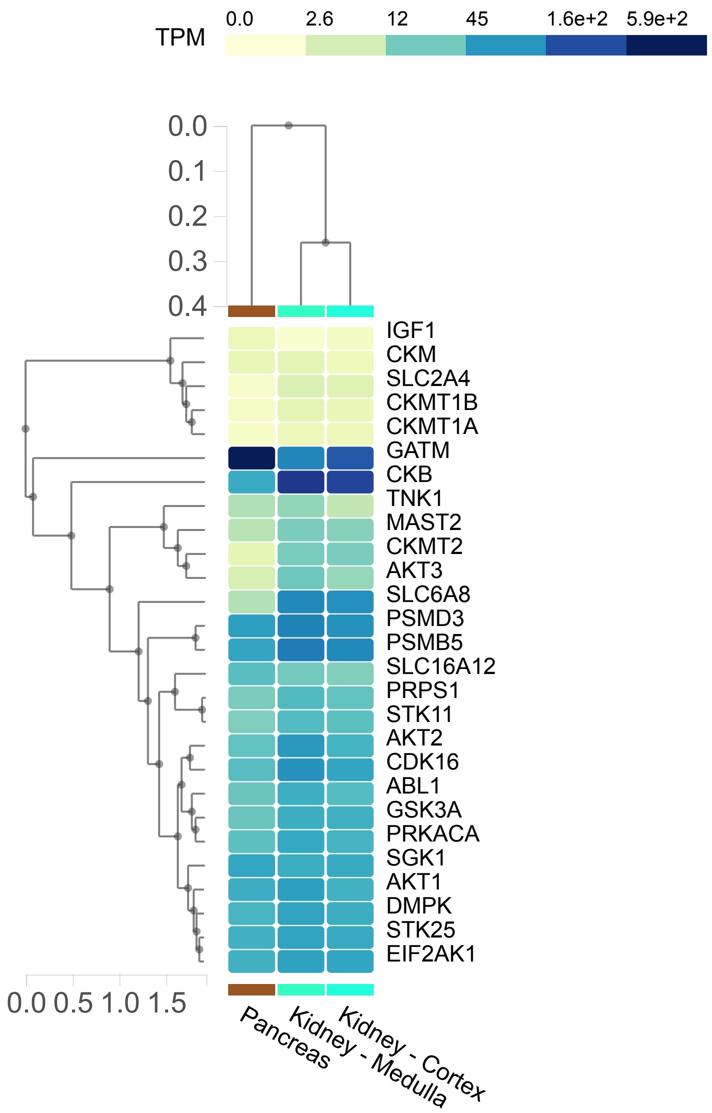
Term	P-value	adjusted P-value	OddsRatio
STK25 human kinase ARCHS4 coexpression	9.992190247741576E-5	0.008343478856864216	20.5350717079530€
CDK16 human kinase ARCHS4 coexpression	9.992190247741576E-5	0.008343478856864216	20.5350717079530€
DMPK human kinase ARCHS4 coexpression	0.001925573805929655	0.024736217353096338	14.2521718146718‡
MAST2 human kinase ARCHS4 coexpression	0.001925573805929655	0.024736217353096338	14.2521718146718‡
PRKACA human kinase ARCHS4 coexpression	0.001925573805929655	0.024736217353096338	14.2521718146718‡
ABL1 human kinase ARCHS4 coexpression	0.001925573805929655	0.024736217353096338	14.2521718146718‡
STK11 human kinase ARCHS4 coexpression	0.001925573805929655	0.024736217353096338	14.2521718146718‡

Term	P-value	adjusted P-value	OddsRatio
TNK1 human kinase ARCHS4 coexpression	0.001925573805929655	0.024736217353096338	14.2521718146718[1]
EIF2AK1 human kinase ARCHS4 coexpression	0.001925573805929655	0.024736217353096338	14.2521718146718[1]
GSK3A human kinase ARCHS4 coexpression	0.001925573805929655	0.024736217353096338	14.2521718146718[1]

Using these kinases symbols, together with the starting gene list, we collected the general tissue expression profiles from the GTex Portal. Interestingly, the expression of these genes in kidney and pancreas cluster together.



GTex expression profiles for the creatine-related genes together with kinases - All tissues.

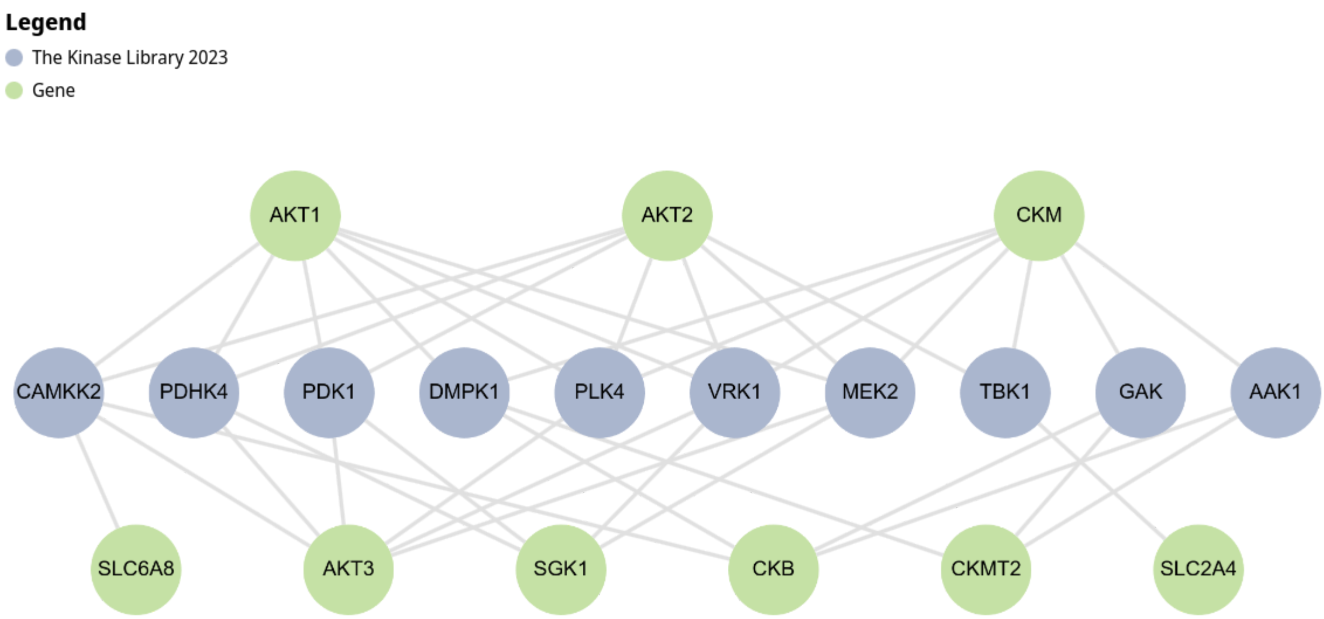


GTex expression profiles for the creatine-related genes together with kinases - Kidney and Pancreas tissues.

We also searched against the dataset Kinase Library 2023. The results are described in the file The_Kinase_Library_2023_table.txt. In this new results we observed additional kinases directly related to proteins of the creatine metabolism, such as CKB, CKM and CKMT1.

Term	P-value	adjusted P-value	OddsRatio
CAMKK2	5.125942179754265E-4	0.022482293685173876	9.602888086642599 72.75173004896666
MEK2	5.651178823731518E-4	0.022482293685173876	9.390459363957596
VRK1	6.548240879176856E-4	0.022482293685173876	9.077347016343596
PDHK4	0.0038968095151162767	0.08545734549852911	7.46552562481766
PLK4	0.004148414830025685	0.08545734549852911	7.3303392259914
PDK1	0.005872262249694354	0.08847744264005385	6.616424116424116
DMPK1	0.006013030082333757	0.08847744264005385	6.56995353639649
AAK1	0.023617828209109744	0.1592957450548465	5.564488143435512
TBK1	0.024624843228191095	0.1592957450548465	5.472396129766648
GAK	0.025309233574566194	0.1592957450548465	5.412614980289093
NEK7	0.025395516635510477	0.1592957450548465	5.405230596175478
PBK	0.025481962898088663	0.1592957450548465	5.3978655682456465
BIKE	0.0261794093655069	0.1592957450548465	5.339633129516398
MASTL	0.026355403573110305	0.1592957450548465	5.325263352430235
VRK2	0.02796874656539955	0.1592957450548465	5.199205345855156

The next figure describes the network obtained with the top 10 rated results:



Kinase pertubartions from GEO

From these datasets we identified the genes from our list that up or downregulated in GEO datasets that have kinase perturbations, like mutants lines, overexpression, etc.

The kinase-perturbation upregulated associated genes were:

Kinase/perturbation	GEO Entry	P-value	adjusted P-value
IRAK4 defectivemutant 200	GSE6789	1.0121923535514024E-4	0.007996319593056078
HIPK2 knockout 171	GSE39253	0.001944001714153394	0.051192045139372706
MAP2K4 knockdown 62	GSE19091	0.001944001714153394	0.051192045139372706
AKT1 activemutant 10	GDS2304	0.026291292253072487	0.14835800628519477
AKT1 activemutant 9	GDS2304	0.026291292253072487	0.14835800628519477
HIPK2 knockdown 106	GSE27869	0.026291292253072487	0.14835800628519477
EGFR drugactivation 19	GDS2146	0.026291292253072487	0.14835800628519477
FGFR1 activemutant 58	GSE17916	0.026291292253072487	0.14835800628519477
MET knockout 247	GDS3148	0.026291292253072487	0.14835800628519477
ROCK2 knockdown 157	GSE34769	0.026291292253072487	0.14835800628519477

And the downregulated ones were:

Kinase/perturbation	GEO Entry	P-value	adjusted P-value
AKT1 activemutant 9	GDS2304	1.0121923535514024E-4	0.008806073475897201
KSR1 knockout 126	GSE28228	0.001944001714153394	0.08456407456567264
MYLK knockdown 48	GSE14525	0.026291292253072487	0.15248949506782042
TGFBR2 knockout 295	GSE45968	0.026291292253072487	0.15248949506782042
TRIM28 knockout 302	GSE32224	0.026291292253072487	0.15248949506782042
PRKACB knockdown 91	GSE27869	0.026291292253072487	0.15248949506782042
PLK2 knockdown 87	GSE27869	0.026291292253072487	0.15248949506782042
MAP2K1 druginhibition 172	GSE39984	0.026291292253072487	0.15248949506782042
WEE1 druginhibition 307	GSE38972	0.026291292253072487	0.15248949506782042
PIK3CA knockdown 182	GSE46869	0.026291292253072487	0.15248949506782042

Regulation by phosphatases

The dataset used for the evaluation of possible phosphatases-mediated regulation was Phosphatase Substrates from DEPOD. All of the retrieved phosphatases relate to the regulation of the genes from AKT family (AKT1, AKT2 and AKT3) associated to creatine metabolism.

Term	P-value	adjusted P-value	OddsRatio
PHLPP1	1.0183823146040292E-8	2.545955786510073E-8	1427.142857142857
PHLPP2	1.0183823146040292E-8	2.545955786510073E-8	1427.142857142857
PPP2CA	0.002493640221385313	0.004156067035642188	30.848062015503874
ACP1	0.012678709933633063	0.01584838741704133	89.14732142857143
PPP1CA	0.0384036388314902	0.03840363883149023	27.691666666666666

Resultado enrichr-KG

The subnetwork shows the following associations:

- **From Gene Ontology:** PSMB5, and PSMD3 belong to the biological process regulation of cellular amino acid metabolic process (GO:0006521). PSMB5, and PSMD3 belong to the biological process regulation of cellular ketone metabolic process (GO:0010565). SLC6A8, and GATM belong to the biological process creatine metabolic process (GO:0006600). PSMB5, and PSMD3 belong to the biological process regulation of cellular amine metabolic process (GO:0033238). PSMB5, and PSMD3 belong to the biological process negative regulation of cell cycle G2/M phase transition (GO:1902750).
- **From KEGG**: The gene products PSMB5, and PSMD3 are members of the KEGG pathway Proteasome. The gene products PSMB5, and PSMD3 are members of the KEGG pathway Huntington disease. The gene products PSMB5, and PSMD3 are members of the KEGG pathway Prion disease. The gene products PSMB5, and PSMD3 are members of the KEGG pathway Parkinson disease. The gene products PSMB5, and PSMD3 are members of the KEGG pathway Spinocerebellar ataxia.
- **From Jensen lab:** The disease Arts syndrome is associated with the gene PRPS1. The disease Intellectual disability is associated with the genes SLC6A8, and PRPS1. The disease Gyrate atrophy is associated with the gene GATM. The disease Purine nucleoside phosphorylase deficiency is associated with the gene PRPS1. The disease AGAT deficiency is associated with the gene SLC6A8.
- **From DisGeNET:** The disease Mammary Carcinoma, Animal is associated with the following genes: SGK1, and GATM. The disease Drug Resistant Epilepsy is associated with the following genes: SLC6A8, and SGK1. The disease Congenital pes cavus is associated with the following genes: SLC6A8, and PRPS1. The disease Neonatal Hypotonia is associated with the following genes: SLC6A8, and PRPS1. The disease Creatine deficiency, X-linked is associated with the following genes: SLC6A8, and GATM.